THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE UNIT TEST SET B

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. Oxidation is: 1

(a) gain of oxygen or loss of hydrogen **(b)** gain of hydrogen
(c) gain of electrons **(d)** loss of oxygen

2. The blue colour of copper sulphate solution fades when an iron nail is dipped in it because: 1

(a) water evaporates **(b)** copper displaces iron
(c) iron displaces copper, forming $FeSO_4$ **(d)** the nail dissolves completely

3. Chips manufacturers flush packets with a gas to prevent rancidity. The gas is: 1

(a) hydrogen **(b)** nitrogen **(c)** oxygen **(d)** carbon dioxide

4. The chemical name of bleaching powder is: 1

(a) sodium hypochlorite **(b)** calcium oxychloride
(c) calcium chloride **(d)** calcium carbonate

5. The salt formed from a strong acid and a strong base has pH: 1

(a) less than 7 **(b)** 7 **(c)** more than 7 **(d)** 0

6. A student states: “All bases are alkalis.” The correct response is: 1

(a) Correct
(b) Incorrect — only water-soluble bases are alkalis
(c) Correct for metal oxides
(d) Cannot be determined

7. Which of these displaces copper from copper sulphate solution? 1

(a) gold **(b)** mercury **(c)** zinc **(d)** silver

8. Ionic compounds generally have: 1

(a) no electrical conductivity in molten state
(b) a liquid state at room temperature
(c) low boiling points
(d) high melting points

9. The property of metals by which they can be beaten into thin sheets is: 1

(a) lustre **(b)** sonority **(c)** malleability **(d)** ductility

10. The addition of hydrogen to vegetable oils in the presence of nickel is called: 1

(a) oxidation **(b)** hydrogenation **(c)** saponification **(d)** substitution

11. The functional group present in ethanol is: 1

(a) –OH **(b)** –COOH **(c)** –CHO **(d)** C=O

12. The molecular formula of the fourth member of the alkene series is: 1

(a) C_4H_{10} **(b)** C_5H_{10} **(c)** C_5H_{12} **(d)** C_4H_8

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 10–15 words each.

13. A student says silver chloride turns grey in sunlight due to oxidation of silver. Justify or refute. 2

14. A student says that alcohol and glucose solutions are acidic because they contain hydrogen. Justify or refute. 2

15. What is an alloy? Give one example with its constituents. 2

16. Why is sodium kept immersed in kerosene oil? 2

17. Why does carbon form covalent bonds rather than ionic bonds? 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 30–40 words each.

18. Calcium oxide reacts vigorously with water. 3

(i) Name the product.
(ii) Name the type of reaction.
(iii) Why do walls whitewashed with it become shiny after a few days?

19. Five solutions A–E have pH 4, 1, 11, 7 and 9. Identify 3

(i) neutral
(ii) strongly alkaline
(iii) strongly acidic
(iv) weakly acidic
(v) weakly alkaline. Arrange them in increasing order of H^+ concentration.

20. Write equations for the reactions of 3

(i) iron with steam
(ii) calcium with water
(iii) potassium with water.

21. What are isomers? Draw the structures of the two isomers of butane. 3

22. What do you observe when ferrous sulphate crystals are heated? Write the equation and name the type of reaction. 3

OR

Write balanced equations with state symbols: 3

(i) barium chloride and sodium sulphate solutions react
(ii) sodium hydroxide reacts with hydrochloric acid
(iii) zinc reacts with silver nitrate solution.

23. What is baking powder? How does it make cakes soft and spongy? Why does a cake made with only baking soda taste bitter? 3

OR

Fresh milk has pH 6. How will the pH change as it turns into curd? Why does a milkman add a very small amount of baking soda to fresh milk? 3

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SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. Identify the type of each reaction: **5**

(a) $2\,KClO_3 \longrightarrow 2\,KCl + 3\,O_2$

(b) $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$

(c) $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$

(d) $2\,Mg + O_2 \longrightarrow 2\,MgO$

(e) $ZnO + C \longrightarrow Zn + CO$

OR

(i) Define corrosion. **5**

(ii) Describe an activity with three test tubes to show that both air and water are needed for rusting.

(iii) Give one method to prevent rusting.

25. (i) Explain how mercury is obtained from cinnabar and copper from copper glance, with equations. **5**

(ii) Why are these metals extracted by heating alone?

OR

(i) What is electrolytic refining? **5**

(ii) Draw/describe the set-up for refining copper.

(iii) What are the anode, cathode and electrolyte?

(iv) Where does anode mud collect?

END OF QUESTION PAPER — Total: 50 Marks

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